

Project Design Phase-II

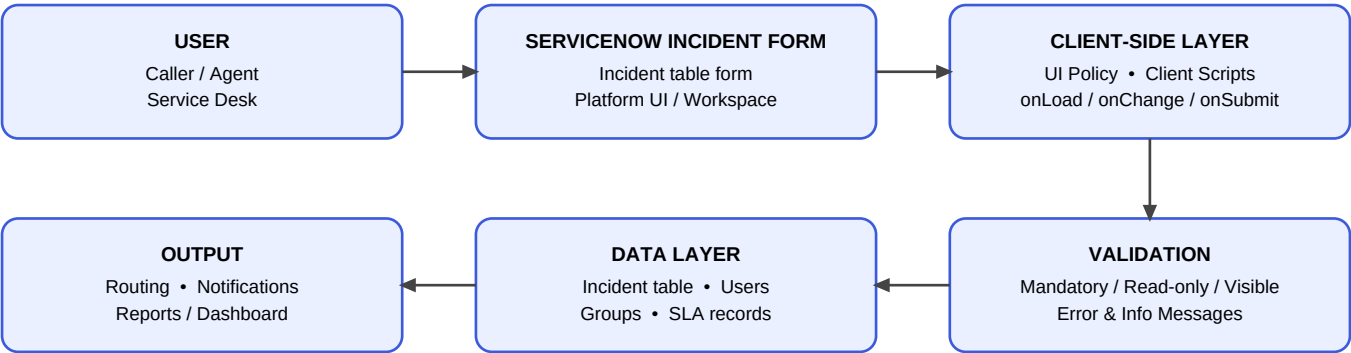
Technology Stack (Architecture & Stack)

|               |                                                |
|---------------|------------------------------------------------|
| Date          | 30 September 2026                              |
| Team ID       | SWTID-2026-9644                                |
| Project Name  | Implement Client Script & UI Policy (Incident) |
| Team Size     | 3                                              |
| Team Leader   | Vairavakarumari A                              |
| Team Members  | Abinaya K; Kayathri Devi B                     |
| Maximum Marks | 4 Marks                                        |

Technical Architecture:

**Project objective:** Enforce data integrity on Incident records in ServiceNow. The application lets callers and agents enter Incidents on a standard form where UI Policies control field behavior, Client Scripts auto-populate and validate data, and invalid submissions are blocked with clear messages before the record is saved.

**Architecture:** User -> ServiceNow Incident Form -> UI Policies / Client Scripts -> Validation & Messages -> Incident Table -> Routing / Notifications / Reports. The solution is deployed on the ServiceNow cloud platform.



Technology Stack / Architecture Components

| S.No. | Component           | Description                                                                               | Technology                                                       |
|-------|---------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 1     | User Interface      | Caller/agent-facing Incident form in ServiceNow where UI Policies and Client Scripts run. | ServiceNow Incident Form (Platform UI / Workspace), Form Layout  |
| 2     | Application Logic-1 | Conditional field behavior: mandatory, visible and read-only rules based on field values. | UI Policy, UI Policy Actions                                     |
| 3     | Application Logic-2 | Client-side logic for defaults, auto-population and save-time validation.                 | Client Scripts (onLoad, onChange, onSubmit), g_form, g_user APIs |
| 4     | Application Logic-3 | Optional server-side enforcement for records that bypass the form (imports, APIs).        | Data Policies, Business Rules (optional)                         |
| 5     | Database            | Stores Incident records, users, groups and related data.                                  | ServiceNow platform database / Incident table                    |
| 6     | Cloud Database      | Managed cloud data layer provided by the ServiceNow platform.                             | ServiceNow Cloud Platform                                        |
| 7     | File Storage        | Optional storage for screenshots, logs or supporting attachments on Incidents.            | ServiceNow Attachments / platform file storage                   |

## Technology Stack (Architecture & Stack) — Continued

| S.No. | Component                       | Description                                                                                                             | Technology                                              |
|-------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 8     | External API-1                  | No mandatory third-party API is required for the core solution; integration can be added for monitoring or email tools. | ServiceNow REST API / IntegrationHub, if required       |
| 9     | External API-2                  | Optional integration with monitoring, email or other ticketing systems that create Incidents.                           | REST / SOAP APIs, IntegrationHub, Inbound Email Actions |
| 10    | Machine Learning Model          | Not required for client-side validation of Incident records.                                                            | Not applicable                                          |
| 11    | Infrastructure (Server / Cloud) | Application is hosted and operated on the ServiceNow cloud platform; no local application server is required.           | ServiceNow Cloud / SaaS platform                        |

**Table-2: Application Characteristics**

| S.No. | Characteristics          | Description                                                                                                                                           | Technology                                        |
|-------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| 1     | Open-Source Frameworks   | The core solution uses the ServiceNow platform rather than a separate open-source framework. JavaScript is used for supported client-side scripting.  | JavaScript / ServiceNow Client APIs               |
| 2     | Security Implementations | Role-based access, ACLs, authentication and authorization; client-side rules are complemented by server-side controls such as Data Policies and ACLs. | ServiceNow ACLs, Roles, Authentication, HTTPS/TLS |
| 3     | Scalable Architecture    | Cloud-based SaaS architecture supports many users; rules can be extended to more fields and tables without maintaining local servers.                 | ServiceNow Cloud Platform, UI Policies            |
| 4     | Availability             | The application runs on the ServiceNow cloud service, reducing dependency on local infrastructure and enabling centralized access.                    | ServiceNow Cloud / SaaS                           |

## Core Automated Flow

|                           |                                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------|
| <b>1. Open Form</b>       | Caller/agent opens the Incident form; an onLoad Client Script sets default values.              |
| <b>2. Apply UI Policy</b> | Fields become mandatory, visible or read-only based on State, Category or Priority.             |
| <b>3. Auto-Populate</b>   | onChange Client Scripts fill dependent fields when a value changes.                             |
| <b>4. Validate</b>        | onSubmit Client Script checks the data and blocks the save with a message if rules are not met. |
| <b>5. Save</b>            | A valid Incident is stored in the Incident table and routed to the resolver group.              |
| <b>6. Closure</b>         | Close code and close notes are required on resolution; the Incident is closed after completion. |

## Technology Stack (Architecture & Stack) — Application Performance

| S.No. | Characteristics | Description                                                                                                                                                                                                                                                                     | Technology                                              |
|-------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 5     | Performance     | Performance is improved by running checks instantly in the browser, using UI Policies instead of scripts where possible, keeping Client Scripts lightweight (avoiding server calls such as client-side GlideRecord), and handling heavy logic on the server only when required. | UI Policy, Client Scripts, g_form API, ServiceNow Cloud |

## Project Team

| Role        | Name              | Team ID         |
|-------------|-------------------|-----------------|
| Team Leader | Vairavakarumari A | SWTID-2026-9644 |
| Team Member | Abinaya K         | SWTID-2026-9644 |
| Team Member | Kayathri Devi B   | SWTID-2026-9644 |

## References

- ServiceNow University — ServiceNow learning platform and project environment.
- ServiceNow Product Documentation — UI Policies, Client Scripts, Data Policies, ACLs and platform architecture.
- ServiceNow Developer Documentation — Client-side JavaScript APIs (g\_form, g\_user) and application development.

**Note:** The architecture is tailored to the project scope "Implement Client Script & UI Policy (Incident)." Optional external integrations and server-side controls are shown as extensibility points; the core solution can be implemented entirely within ServiceNow.